Abstract

However, there are many problems that many farmers face while receiving agricultural information on time and conveniently because of language barriers, low levels of digital literacy, and unavailable agricultural experts when needed. Therefore, a project idea was formed – **Voice-Based Bengali Crop Advisory AI Agent**, which is intended to help Bengali-speaking farmers in West Bengal.

Farmers would have the opportunity to communicate with the AI agent using only their voice in Bengali. The system would consist of Speech-to-Text technology that converts speech into text, an AI agent that understands the user's request, determines what kind of information should be retrieved, and uses an appropriate tool or source of knowledge to give a meaningful answer. An agricultural knowledge base would be included in the project, which would contain data about different crops, diseases, pests, fertilizers, irrigation, methods of cultivation, prevention, etc. There could also be a possibility to incorporate a Retrieval-Augmented Generation (RAG) to retrieve credible agricultural information and generate a relevant response to the request.

Unlike the chatbot, the proposed AI agent can work with different external tools like weather service, an agricultural knowledge base, crop-disease detection model. Thus, the agent could combine the farmer's crop-related question with the information about current weather conditions and give an advisory answer after retrieving the necessary information from the agricultural knowledge base. In case of the advanced version, farmers also can send the images of affected crops or leaves. These images will be analyzed by a computer vision model to identify the possible diseases. In the end, the generated recommendation will be converted into Bengali speech with the help of Text-to-Speech technology.

Thus, the main goal of the presented project idea is not to replace agricultural experts but to provide 24/7 first-level agricultural assistance and help Bengali-speaking farmers make proper decisions. In complicated or uncertain situations, there will be a suggestion of consulting an agricultural expert. With the combination of Bengali voice interaction, AI agents, RAG, agricultural knowledge, external tools, weather information, and crop-disease image analysis, the mentioned AI agent will be able to give a proper response.